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Ashad Ahmad

Education

July 2024–Present **Physial Research Laboratory, Ahmedabad**,
PhD, Astronomy and Astrophysics.

July 2022–July 2024 **Indian Institute of Technology, Indore**,
Master of Science, Astronomy.
CPI – 9.38 out of 10

July 2019–Sep 2022 **Karim City College, Jamshedpur**,
Bachelor of Science, Physics (Honours).
Overall Percentage – 85.92%, CGPA – 9.19 out of 10

April 2016–May 2018 **Delhi Public School, Jamshedpur**,
All India Senior School Certificate Examination (AISSCE), Class 12.
Percentage – 78%

April 2008–April 2016 **Modern International School, Al-Hasa, K.S.A**,
Matriculation Examination, Class 10.
CGPA – 10.0

Experience

Mar 2022–Jul 2022 **Research Intern**, DTU Space, Technical University Of Denmark, Denmark.
- Studying X-ray pulsars during it's outbursts and low-luminosity states
- Working under the guidance of Dr. Gaurava Kumar Jaisawal.

Feb 2022–Sept 2022 **Summer Research Intern - Radio Astronomy**, Naxxatra Science.
- Studying the Very Long Baseline Interferometry (VLBI) used by Event Horizon Telescope (EHT)
- Learning Convolution, Deconvolution and Fourier Transform
- Learning to use the CLEAN algorithm on simulated data

Sept 2021–Ongoing **Subject Matter Expert**, Chegg Inc.
- Working as a Subject Matter Expert in Advanced Physics at Chegg India.
- Explaining the concept along with the process and steps with utmost clarity to students.
- Have been maintaining an overall 100% C.F Score

Jun 2021–Aug 2021 **Astrophysics Research Fellowship Program**, SARSTEM.
- Completed a project titled 'Gravitational Wave Data Analysis - Parameter Estimation using Bayesian Analysis' under the supervision of Mr. Edoardo Altamura, Ph.D. Candidate at University of Manchester, U.K.
- Showed excellent pace of learning and performed data analysis
- Worked on the project for 2.5-months and achieved the desired results

Thesis and Projects

May 2023–May 2024 ***Masters thesis* - Understanding the formation of peculiar features in AGN Jets at feedback scales**, *Dr. Bhargav Vaidya, Indian Institute of Technology, Indore*.
- Learning the kinematics, dynamics, morphology, and various emission mechanisms of Active Galaxies and AGN (Active Galactic Nuclei) Jets
- Performing 3D Magnetohydrodynamical (MHD) simulations using the freely available `PLUTO` Code

Jan 2023–May 2023 ***Minor Semester Project* - AA608 - Astrostatistics**, *Dr. Suman Majumdar and Prof. Abhirup Datta, Indian Institute of Technology, Indore*.
- Wrote the code for Metropolis-Hastings algorithm of Markov Chain Monte Carlo to estimate cosmological parameters using Supernova Ia data
- Performed a Gelman-Rubin Convergence Test to my code

Jan 2023–May 2023 ***Minor Semester Project* - AA609 - Computational Methods in Astronomy**, *Dr. Bhargav Vaidya and Dr. Suman Majumdar, Indian Institute of Technology, Indore*.

- Numerical Integration using various ODE algorithms, Stiff problems, One Dimensional Advection Equation
- Flux Limiters - Shock Capturing Schemes, 1D Isothermal Shock Tube
- Convolution, N-Body Simulation

Mar 2022–Jul 2022 **X-Ray Pulsars during outbursts and low-luminosity states**, *Dr. Gaurava K. Jaisawal, Technical University of Denmark*.

- Learned and performed spectral and temporal analysis of various sources
- Created scripts to automate the process of spectral analysis
- Used data available from various X-ray telescopes

Jun 2021–Aug 2021 **Gravitational Wave Data Analysis - Parameter Estimation using Bayesian Analysis**, *Edoardo Altamura, SARSTEM*.

- Studied The General Theory of Relativity with its applications (Black Holes, Gravitational Waves, and LIGO detectors).
- Analysed data from three LIGO/Virgo observation runs to perform parameter estimation and compare the published results with the estimated data.

Schools and Symposiums

July 2023 **Workshop on Computational Magneto-hydrodynamics (MHD) with PLUTO**, .

- Overview of Hardware/Architecture of CPU and GPU, Parallel Programming using Message Passing Interface (MPI)
- Theoretical and Computational aspects of MHD, Numerical Methods, Gudonov Scheme, Reimann Problem
- Basics of GRMHD (General Relativistic MHD) and Particle-in-Cell (PIC) method
- Several hands-on sessions involving simulation using PLUTO and analyzing data using *VisIt* and *pyPLUTO*.

Feb 2022 **X-ray Astronomy School**, .

The topics covered in the school are:

- Introduction to X-Ray Astronomy
- X-Ray Telescopes and Observatories (NUSTAR, XMM-Newton, Swift Observatory, Chandra X-Ray Observatory)
- Hands-on Training with actual data using HEASOFT and FTOOLS such as XSPEC, XSELECT etc

Feb 2022 **SOKENDAI Asian Winter School, Astronomy**, NATIONAL ASTRONOMICAL OBSERVATORY OF JAPAN (NAOJ).

The topics covered in the school are:

- Galaxy Evolution, Galaxy Clusters, Planet Formation and The Sun
- Galactic Archaeology and Astro-chemistry
- Gravitational Wave Astronomy, Black Holes and Active Galactic Nuclei

Key Courses

Computational Methods in Astronomy.

- Learned basics of computational physics particularly, solving Partial Differential Equations (PDEs)
- Learned Computational fluid dynamics and wrote various Python codes to solve the Hydrodynamics equation
- Learned *N-Body Simulations* and performed statistical analysis on simulated data

Astrostatistics.

- Learned Applied Probability and Statistics
- Wrote Python codes performing *parameter estimation using Markov Chain Monte Carlo sampling algorithm*

Astrophysical Fluid Dynamics.

- Learned the basics of Hydrodynamics (HD) and Magneto-Hydrodynamics (MHD) equations
- Learned about various instabilities and applications of HD and MHD in astrophysical scenarios

Galactic & Extra-galactic Astronomy.

- Understood the Milky Way and other galaxies' structures, including spiral arms, bulges, and halos.
- Studied high-energy phenomena like quasars and black holes at the centers of galaxies.
- Learned the movement and interactions of stars, gas, and dark matter within galaxies.

Achievements

Mar 2024 **Joint Entrance Screening Test (JEST), Physics**.

- Qualified the JEST Entrance Test (a National Eligibility Test for PhD in India) with a percentile of 91.28
- Obtained an All India Rank (AIR) *421*

Mar 2024 **Graduate Aptitude Test in Engineering (GATE), Physics**.
- Qualified the GATE Exam (a National Eligibility Test for PhD in India) with an overall score of **531** out of 1000
- Obtained an All India Rank (AIR) *675*

Oct 2023 **International English Language Testing System (IELTS)**.
- Scored an **Overall Band** of **7.5**
- Listening (L): 8.0, Reading (R): 8.5, Writing (W): 7.0, Speaking (S): 6.5

Mar 2022 **Joint Admission test for Masters (Physics), IIT-JAM**.
- Qualified the National Entrance Test for Master Degree with an overall score of **55.33** out of 100.
- Obtained an All India Rank (AIR) *407*

2015 **National Level Science Talent Search Examination**, NSTSE.
- Scored an overall of 54 out of 100 marks
- Obtained an All India Rank (AIR) *10246*

Nov 2014 **Unified International English Olympiad**, UIE.
- Achieved an All India Rank (AIR) *158*
- Achieved Zone Rank *1*

2014 **National Level Science Talent Search Examination**, NSTSE.
- Scored an overall of 41 out of 100 marks
- Obtained an All India Rank (AIR) *21479*

Nov 2013 **Unified International English Olympiad**, UIE.
- Achieved an All India Rank (AIR) *1228*
- Achieved Zone Rank *13*

Outreach Activities

Sep 2023 ***100 Hours of Astronomy*** **- International Astronomical Union (IAU)**.
- Presented a talk titled, "*Observational Astronomy - Looking at our universe from different perspectives*"
- Volunteered as a guide in an accompanying stargazing session.

Oct 2022 ***Outreach DAASE*** **- Indian Institute of Technology Indore**.
- Volunteered as a guide in star gazing and Partial Solar Eclipse Events.
- Learned and handled several telescopes involved.

Technical Skills

Languages	Python, C, C++, LaTeX, HTML, TCL Scripting, Shell Scripting, MATLAB
Operating System	Windows, Linux, MacOS
Astronomy Codes	PLUTO, Heasoft, GADGET, OneHaLe
Utilities	Git/Github, Mathematica, VisIt
Communication	English(C1), Hindi(C2), Urdu(B2)

Research Interests

- Active Galaxies
- Relativistic Jets
- Formation and Evolution of Galaxies
- Supermassive Black Holes
- Computational Astrophysics
- Gravitational Waves